

# **“Collaborative Story/Screenplay Writing Platform with Version History”**

**CO5 ASSESSMENT TOOL 2**

*Submitted in the partial fulfilment for the Course of*

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*to the award of the degree of*

**BACHELOR OF ENGINEERING**

**IN**

**Artificial Intelligence & Data Science**

**Submitted by**

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**Under the Supervision**

**of**

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**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

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## DECLARATION

I, **Lithish Kumar R 192424179**, of the **Department of Artificial Intelligence and Data Science**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **“Collaborative Story/Screenplay Writing Platform with Version History”** is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai  
Date: 14/08/2026

Signature of the Students

## 1. Introduction

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Collaborative fiction and screenplay writing increasingly happens across distributed teams of authors, editors, and story consultants who need to write together in real time while preserving the ability to revisit, compare, and recover earlier drafts of scenes, dialogue, and plot structure. General-purpose word processors provide either coarse-grained, hard-to-navigate revision history or shallow “track changes” annotations that are lost once changes are accepted, while software-engineering version control tools such as Git are powerful but built around line-based diffs of source code, not narrative units such as scenes, beats, and character arcs. This literature survey examines the two bodies of research that underpin a Collaborative Story / Screenplay Writing Platform with Version History: (a) real-time collaborative editing (concurrency control techniques such as Operational Transformation and Conflict-Free Replicated Data Types) and (b) version control and history-keeping designed for creative, non-code writing. The survey synthesises objectives, methods, findings and limitations of six representative works, compares them on common parameters, identifies the research gap relevant to the assigned problem statement, and proposes future directions for a platform purpose-built for collaborative narrative and screenplay authoring.

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## 2. Problem Statement

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Design and evaluate a Collaborative Story / Screenplay Writing Platform with Version History that allows multiple authors to co-write a story or screenplay simultaneously, resolves concurrent edits without data loss, and maintains a complete, browsable, and restorable history of narrative revisions at the level of scenes, dialogue, and structural elements – rather than only at the level of raw text characters or whole-file snapshots.

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## 3. Selected Research Papers – Individual Reviews

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### 3.1 Ellis & Gibbs (1989) and Sun & Ellis (1998) – Operational Transformation

These foundational works introduced Operational Transformation (OT) to allow multiple users to edit a shared document concurrently without locking, by transforming each incoming operation against previously executed concurrent operations so that all replicas converge to the same final state. Sun and Ellis formalised the correctness properties – convergence, causality preservation, and intention preservation – that later OT algorithms are still judged against, and OT went on to become the concurrency-control technique behind large-scale collaborative editors including Google Docs and Google Wave.

### 3.2 CRDT-Based Collaborative Editing (Shapiro et al.; sequence-CRDT literature, 2011–2015)

As an alternative to OT, Conflict-Free Replicated Data Types (CRDTs) such as WOOT, Logoot, Treedoc and RGA guarantee eventual consistency purely through commutative data-structure design, without a transformation function or central server. Comparative evaluations of CRDT-based editors show they are well suited to peer-to-peer and offline-first collaboration, though they typically incur higher metadata overhead than OT and, despite strong theoretical guarantees, have seen more limited adoption in production co-editors.

### 3.3 Ink & Switch – Upwelling: Combining Real-Time Collaboration with Version Control for Writers (2023)

Upwelling is an experimental editor built specifically for professional writers and editors, motivated by interviews showing that writers want creative privacy – the ability to draft without every keystroke being visible – while editors need to follow a document's development closely. Rather than treating real-time co-editing and asynchronous suggestion/review as separate modes, Upwelling unifies them: authors can work on private drafts and merge changes into a shared document on their own schedule, with all edits tracked so they can be highlighted retrospectively. This directly anticipates the hybrid synchronous/asynchronous requirement of a collaborative screenplay platform.

### **3.4 Serman, Karahalios & Cheng – Towards Creative Version Control (2022)**

Through eighteen semi-structured interviews with creative practitioners across writing, design, music and programming, this study found that creators treat past versions of their work as reusable materials – branching from, comparing, and remixing earlier states – in ways that existing, developer-oriented version control systems do not support well. The paper argues that history-keeping tools for creative work must be redesigned around non-linear exploration rather than the linear commit histories familiar from software engineering.

### **3.5 Ruskov & Montanelli – A Systematic Literature Review of Online Collaborative Story Writing (2023)**

This systematic review surveys the landscape of online and crowdsourced collaborative fiction-writing platforms and distils four open research directions: how to recruit suitable collaborators, how to give collaborators awareness of others' contributions, how to provide creative guidance, and what modalities of contribution (sentence, scene, or full-draft level) best support quality outcomes. The review finds that few existing platforms combine strong authorship/revision tracking with mechanisms that actively support narrative coherence across contributors.

### **3.6 Interaction and Outcomes in Collaborative Storytelling Systems (Springer CSCW, 2017)**

This paper proposes a six-dimension analytical framework covering both system features and collaboration outcomes (process and output) for collaborative storytelling systems, validated through an extended-duration controlled field study. The study demonstrates that structural design choices – turn-taking rules, awareness cues, and editing permissions – measurably affect narrative coherence and participant satisfaction, providing evaluation criteria that a new platform's design can be benchmarked against.

## 4. Comparative Literature Survey Table

| Author /                                                                                 | Research                                                                                     | Methodology                                                                                                              | Dataset /                                                                       | Key Findings                                                                                                                                  | Limitations                                                                                                                                  | Research Gap                                                                                                                                          |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ellis & Gibbs (1989); Sun & Ellis (1998)                                                 | Maintaining consistency when multiple users edit shared text concurrently                    | Introduced Operational Transformation (OT); formalised convergence, causality- and intention-preservation properties for | GROVE outline editor; formal proofs and simulation of concurrent edit logs      | OT enables lock-free, low-latency concurrent editing and became the basis of systems such as Google Docs and Google Wave                      | Early OT algorithms suffered from the dOPT puzzle and scalability limits with vector timestamps as user count grows                          | OT was designed for generic plain text, not for narrative units (scenes, dialogue blocks, character arcs) that a story/screenplay platform must track |
| Shapiro et al.; Kleppmann & related CRDT literature (2011–                               | Achieving eventual consistency in distributed collaborative editors without a central server | Conflict-Free Replicated Data Types (CRDT) – WOOT, Logoot, RGA – evaluated for real-time document editing                | Simulated multi-site editing sessions; sequence-CRDT benchmarks                 | CRDTs guarantee convergence without transformation logic and support peer-to-peer/offline collaboration                                       | Higher metadata/storage overhead (tombstones) and limited real-world adoption compared with OT-based co-editors                              | No CRDT study addresses semantic, scene-level merging needed for creative narrative content                                                           |
| Ink & Switch – “Upwelling” project (2023)                                                | Reconciling real-time collaboration with writers’ need for creative privacy and asynchronous | Design and prototype of an experimental editor unifying real-time co-editing with git-like version                       | Qualitative interviews with professional writers/editors; working               | A hybrid model that lets authors work privately and merge drafts on their own terms better matches professional writing workflows than pure   | Prototype-stage system; not evaluated at scale or for screenplay-specific formatting and multi-author fiction                                | Lack of a production-ready platform that combines this hybrid model with story/screenplay-specific structuring                                        |
| Sterman et al. – “Towards Creative Version Control”                                      | Understanding how creative practitioners use past versions of work as materials in their     | 18 semi-structured interviews with practitioners across creative domains (writing,                                       | Qualitative coding of interview transcripts; cross-domain thematic              | Practitioners reuse, compare, and branch from old versions non-linearly; generic software VCS (e.g., Git) does not match                      | Findings are domain-general and not translated into a concrete system design for collaborative                                               | Absence of a version-history interface tailored to non-linear exploration of plot/character                                                           |
| Ruskov & Montanelli – Systematic Literature Review of Online Collaborative Story Writing | Mapping the state of research on online, multi-author collaborative fiction writing          | Systematic literature review of collaborative story-writing platforms and crowdsourcing approaches                       | Corpus of peer-reviewed papers on collaborative/crowdsourced narrative systems  | Identifies four open research directions: collaborator recruitment, collaborator awareness, creative guidance, and modalities of contribution | Review highlights that most platforms lack robust mechanisms for tracking authorship, revision history, and creative guidance simultaneously | No reviewed system integrates fine-grained version history with real-time co-authoring awareness for narrative teams                                  |
| Springer CSCW – “Interaction and Outcomes in Collaborative                               | How system design choices affect process and output quality in collaborative storytelling    | Proposed a six-dimension analytical framework; controlled extended-duration field                                        | Field study logs and a comparative analysis against prior collaborative-fiction | System structure (turn-taking, awareness cues, editing rights) measurably shapes narrative coherence and participant satisfaction             | Framework is descriptive/analytical; does not prescribe a version-control or concurrency-control architecture                                | Need to connect storytelling-system design frameworks with concrete revision-history and conflict-resolution mechanisms                               |

## 5. Critical Analysis and Research Gap

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Across the reviewed literature, two research threads have developed largely in isolation. The concurrency-control literature (OT and CRDTs) solves the technical problem of merging simultaneous edits correctly and efficiently, but treats a document as an undifferentiated stream of characters, with no awareness of narrative structures such as scenes, character dialogue blocks, or screenplay-specific formatting elements (sluglines, action lines, parentheticals). The creative-writing and version-control literature (Upwelling, Sterman et al., the collaborative story-writing review, and the CSCW storytelling framework) identifies exactly the structural and workflow needs – creative privacy, non-linear branching, scene-level awareness, narrative coherence – that a platform must satisfy, but does not couple these needs to a concrete, scalable concurrency-control mechanism.

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The consolidated research gap is therefore: no existing system integrates (i) low-latency, provably convergent real-time concurrency control at the character/operation level with (ii) a version-history and branching model designed around narrative units (scenes, beats, character arcs) rather than raw line diffs, and (iii) screenplay-aware formatting and coherence support for multi-author teams. Closing this gap requires a platform-level design that layers a narrative-aware history and merge model on top of an OT- or CRDT-based real-time editing core.

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### 5.1 Key Limitations Identified

- OT-based systems achieve real-time convergence but do not model narrative semantics, making merges of parallel plot branches meaningless at the character-operation level.
- CRDT-based systems trade off storage/metadata overhead for decentralised convergence, which is costly at the scale of a full-length screenplay revised over months.
- Upwelling demonstrates the hybrid real-time/asynchronous model but has only been validated with general professional writers, not specifically for multi-author screenplay teams with format-specific constraints.
- Creative version-control interviews (Sterman et al.) surface user needs but stop short of a system architecture or evaluation for collaborative (multi-author) settings.
- The collaborative story-writing review and the CSCW storytelling framework both point to a lack of platforms that combine authorship/version tracking with active narrative-coherence support.

## 6. Future Directions and Relevance to the Assigned Problem

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Based on the identified gap, a Collaborative Story / Screenplay Writing Platform with Version History should pursue the following directions, each directly relevant to the assigned problem statement:

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- Adopt a CRDT- or OT-based real-time editing core for low-latency co-authoring, but expose version history at the granularity of scenes and dialogue blocks rather than raw character diffs, drawing on Upwelling's hybrid real-time/asynchronous model.
- Design a non-linear, branch-and-merge visualisation of narrative history – informed by Sterman et al.'s findings on how creative practitioners actually reuse past versions – so authors can compare alternative plot lines or character arcs side by side rather than following a single linear commit log.
- Incorporate screenplay-specific structural awareness (scene headings, character cues, dialogue, action lines) into both the concurrency-control layer and the diff/merge view, addressing the format gap left open by generic OT/CRDT text-editing research.
- Add collaborator-awareness and narrative-coherence support (e.g., highlighting when a concurrent edit affects an established character trait or unresolved plot thread), extending the analytical framework from the CSCW collaborative-storytelling field study into an active, in-editor feature.
- Evaluate the resulting platform using the process/output outcome dimensions proposed in the storytelling-systems framework, plus the four open research directions raised by the systematic review of online collaborative story writing – collaborator recruitment, awareness, creative guidance, and contribution modality.

## 7. Conclusion

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The reviewed literature shows that the building blocks required for a Collaborative Story / Screenplay Writing Platform with Version History already exist in two separate research communities: distributed-systems research has solved real-time concurrency control through OT and CRDTs, while HCI and CSCW research has characterised how creative writers actually want to branch, merge, and review narrative history. No reviewed system, however, combines both at the level of narrative structure. A platform that layers scene-aware, non-linear version history and screenplay-format sensitivity on top of a proven real-time concurrency-control core would directly close this gap and give collaborative writing teams a tool that matches both the technical rigor of software version control and the creative workflow of story and screenplay development.

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## 8. References (IEEE Style)

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- [1] C. A. Ellis and S. J. Gibbs, "Concurrency control in groupware systems," in Proc. ACM SIGMOD Conf. Management of Data, Portland, OR, USA, 1989, pp. 399–407.
  - [2] C. Sun and C. Ellis, "Operational transformation in real-time group editors: Issues, algorithms, and achievements," in Proc. ACM Conf. Computer Supported Cooperative Work (CSCW), Seattle, WA, USA, 1998, pp. 59–68.
  - [3] C. Sun, X. Jia, Y. Zhang, Y. Yang, and D. Chen, "Achieving convergence, causality-preservation, and intention-preservation in real-time cooperative editing systems," ACM Trans. Comput.-Hum. Interact., vol. 5, no. 1, pp. 63–108, Mar. 1998.
  - [4] M. Shapiro, N. Preguiça, C. Baquero, and M. Zawirski, "Conflict-free replicated data types," in Stabilization, Safety, and Security of Distributed Systems, Berlin, Germany: Springer, 2011, pp. 386–400.
  - [5] Ink & Switch, "Upwelling: Combining real-time collaboration with version control for writers," Ink & Switch Research Essay, 2023. [Online].
  - [6] S. Sterman, K. Karahalios, and J. Cheng, "Towards creative version control," Proc. ACM Hum.-Comput. Interact., vol. 6, no. CSCW2, Article 336, 2022.
  - [7] M. Ruskov and S. Montanelli, "A systematic literature review of online collaborative story writing," in Human-Computer Interaction – INTERACT 2023, Cham, Switzerland: Springer, 2023, pp. 65–85.
  - [8] "Interaction and outcomes in collaborative storytelling systems: A framework, a field study, and a model," Comput. Support. Coop. Work (CSCW), vol. 26, pp. 949–992, 2017.
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